

智能结构与智慧港口

Smart Marine Metastructures

浙江大学海洋学院
Ocean college, Zhejiang University

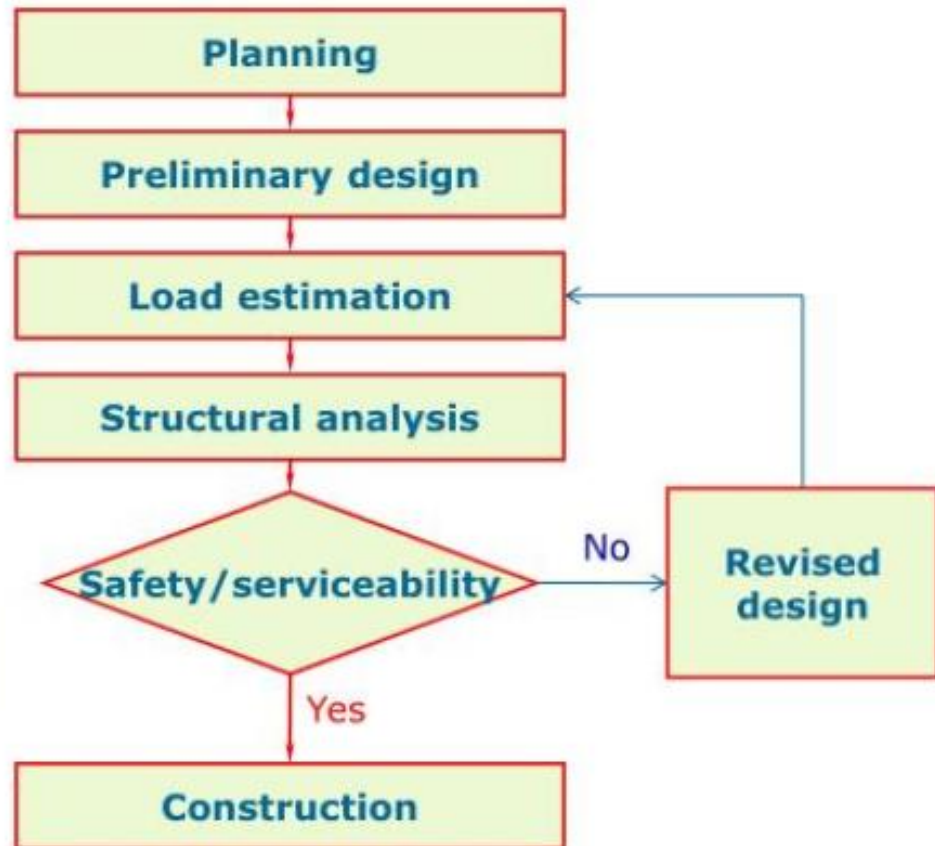


CALENDERS

Week	Subject	Content
1	Introduction and Overview	1. Overview of structural and material analysis 2. Introduction of structural materials and vision for their applications in ocean engineering
2	Advanced Metastructures	1. Structures vs. metastructures 2. Advanced characteristics of metastructures
3	Structural Health Monitoring (SHM)	1. Advent and development of structural health monitoring (SHM) 2. Main SHM technologies
4	SHM in Civil and Ocean Engineering	1. SHM in civil and ocean engineering 2. Challenges and future work
5	Introduction to Intelligent Ports	1. Traditional ports vs. modern intelligent ports 2. Information networks of intelligent ports
6	Equipment and Cases of Intelligent Ports	1. Typical equipment in intelligent ports 2. Case study of intelligent ports
7	Structural Design and SHM in Data Era	1. Vision for smart marine metastructures and intelligent ports 2. Summary of the class
8	Oral Presentation	

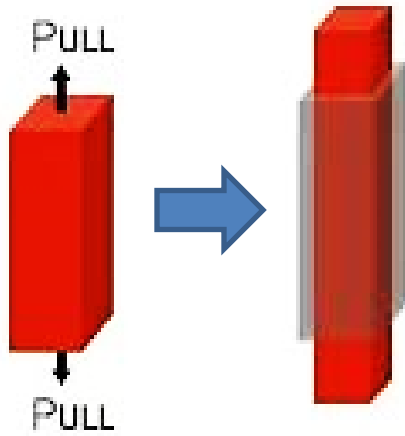
Structural Analysis

What is Structural Analysis?

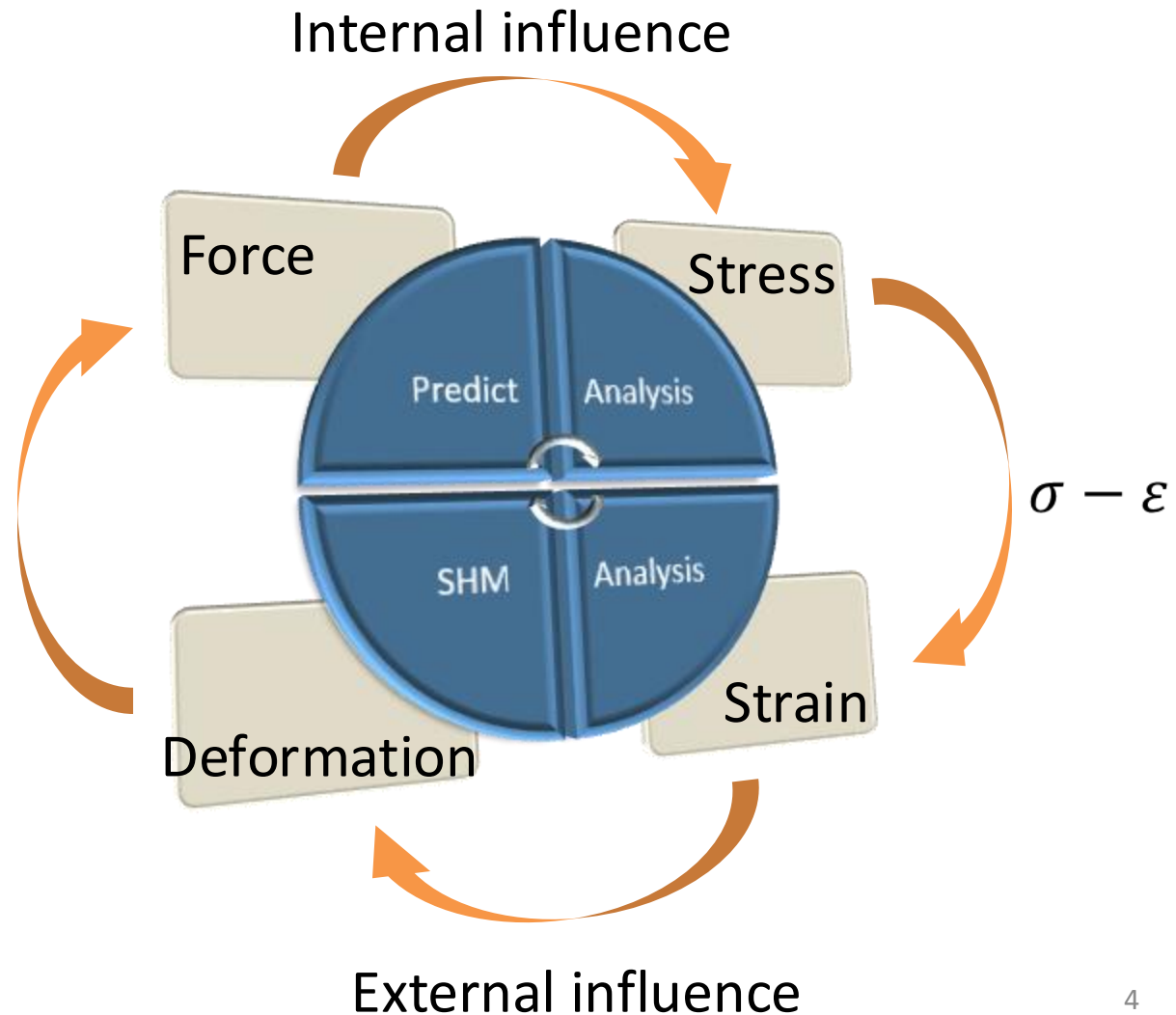


Material Property

Background



Newton's
laws



Metastructures

Young's modulus E

Micro-/nanolattices

- Stretch-dominated octet-truss
- Bend-dominated tetrakaidecahedron

Chiral/anti-chirals

- Trichirals
- Chiral systems — ○ Tetrachirals
- Hexachirals
- Anti-chiral systems — ○ Anti-trichiral
- Anti-tetrachiral

Origami metamaterials

- Miura-ori tessellated pattern
- Non-periodic Ron Resch pattern
- Square twist
- Kirigami

Cellular origami

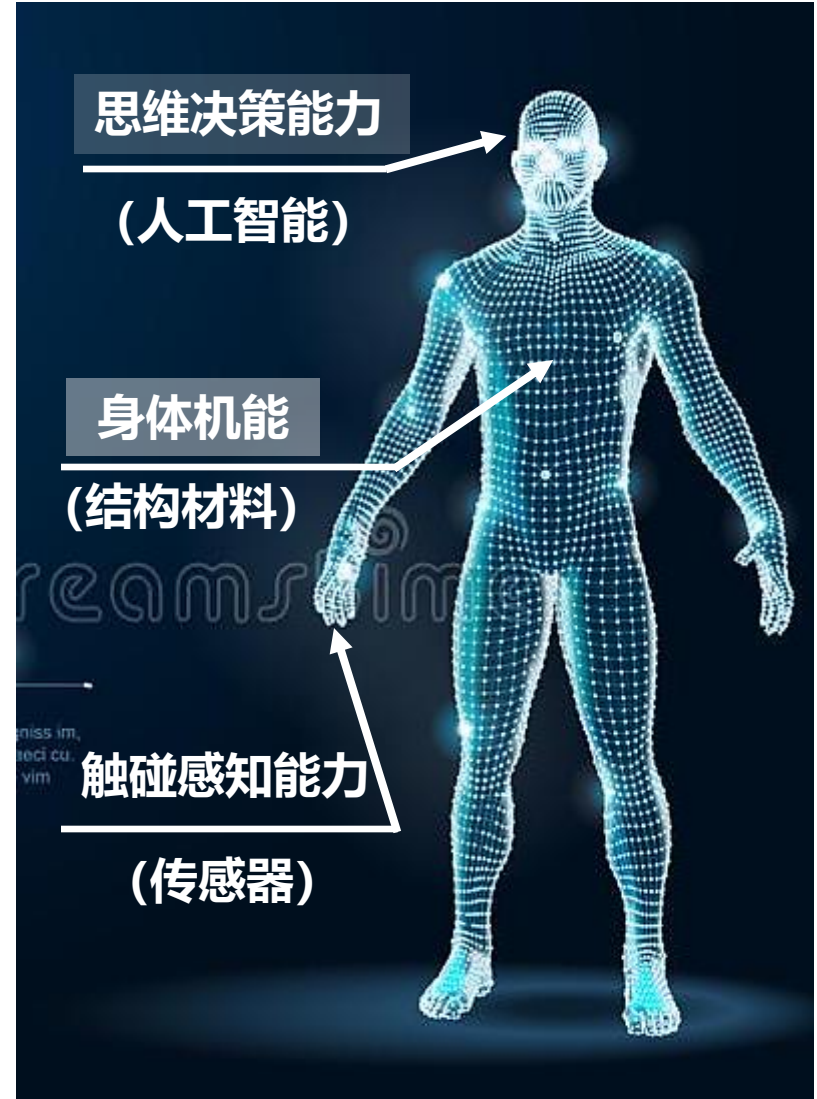
- Stacked Miura-ori
- Interleaved Miura-ori

Pattern transformation

- Holey sheets
- Biholar sheets

《智能结构与智慧港口》

- ❑ What is intelligence?
- ❑ Why is intelligence of interest?
- ❑ How to be intelligent?
- ❑ When is intelligence achievable?



Structural Health Monitoring (SHM)

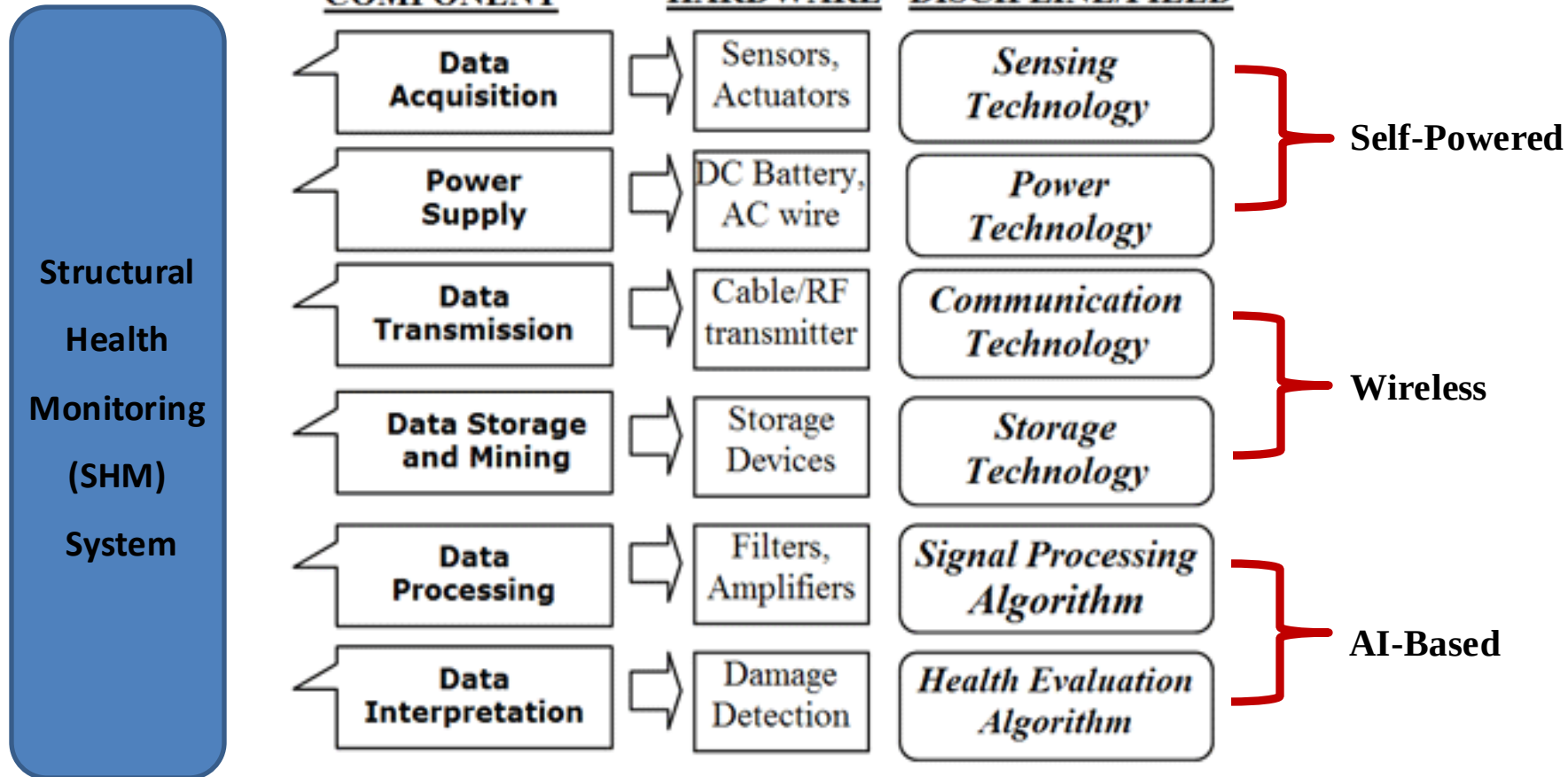
Background



SHM: Classification // Location // Significance

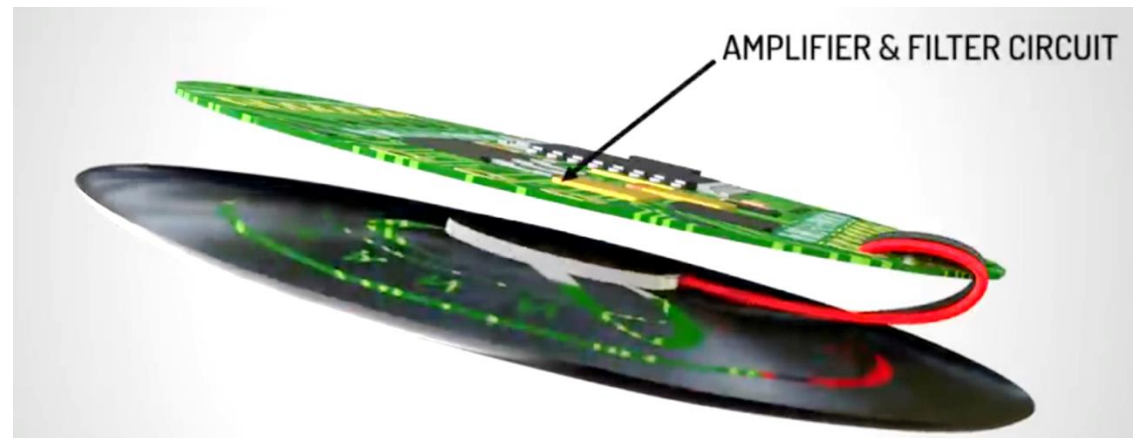
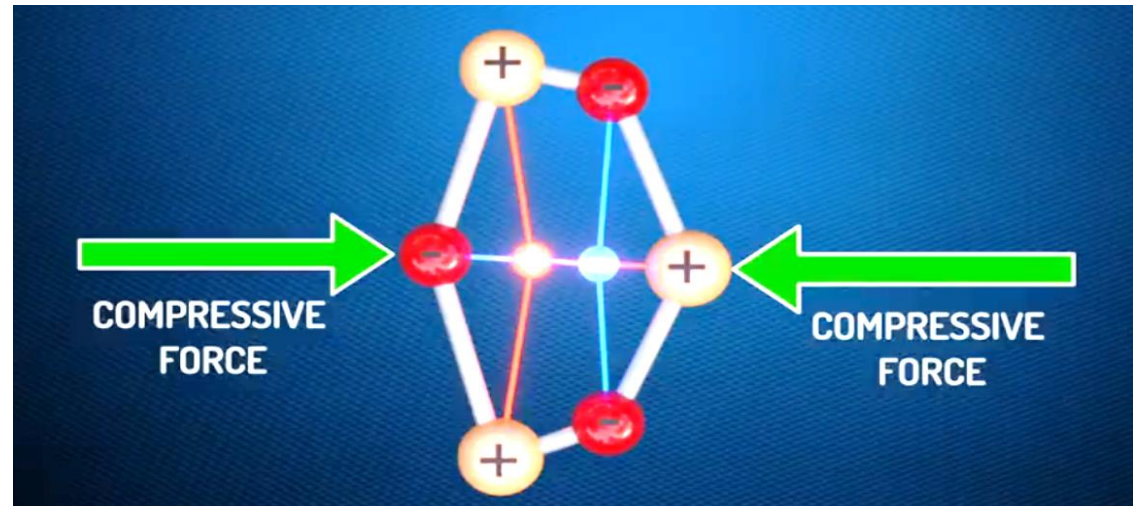
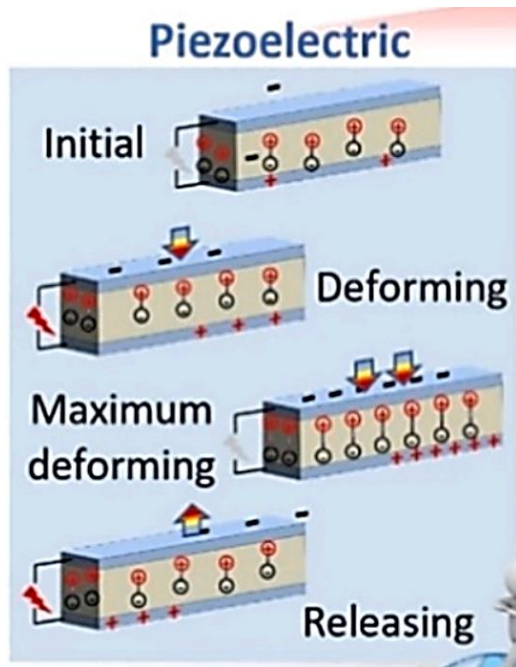
Structural Health Monitoring (SHM)

Background



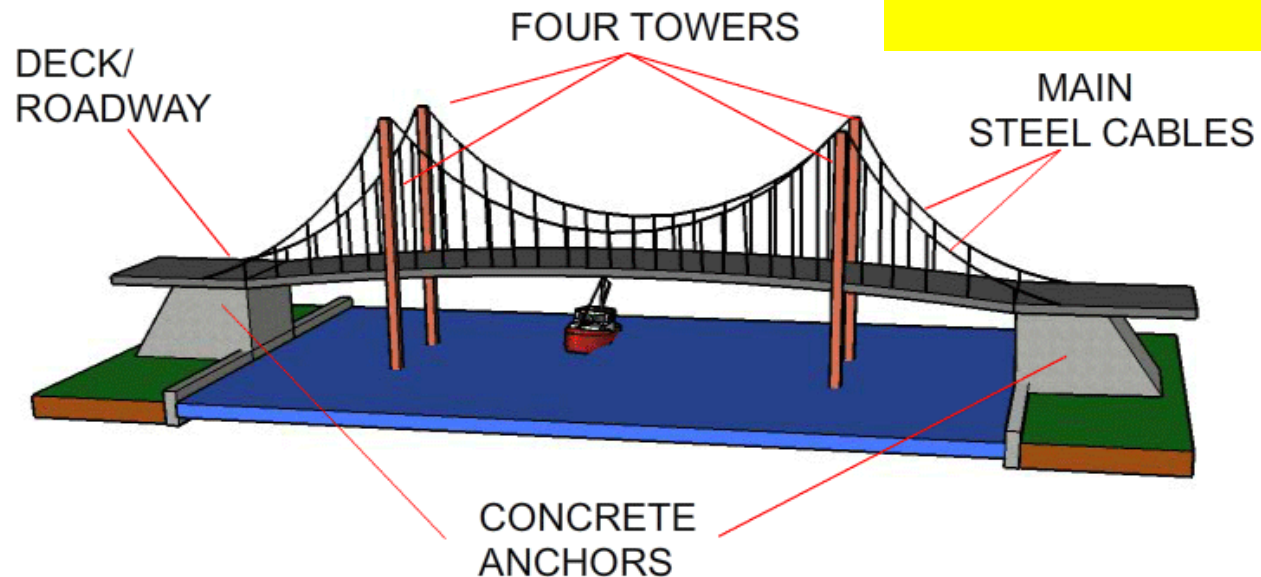
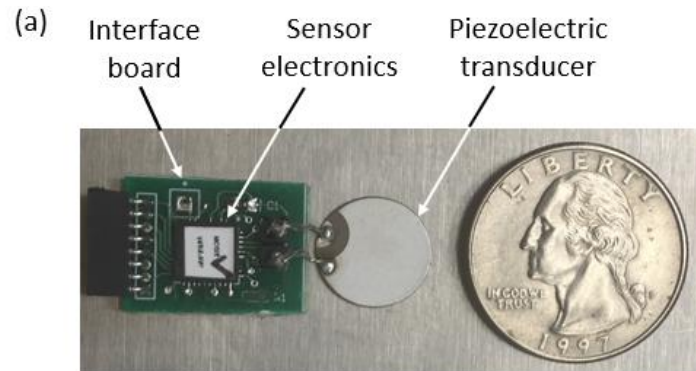
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors



Structural Health Monitoring (SHM)

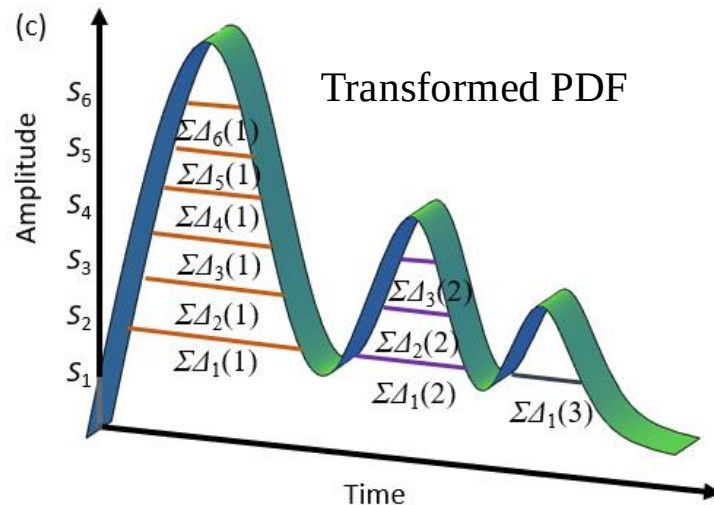
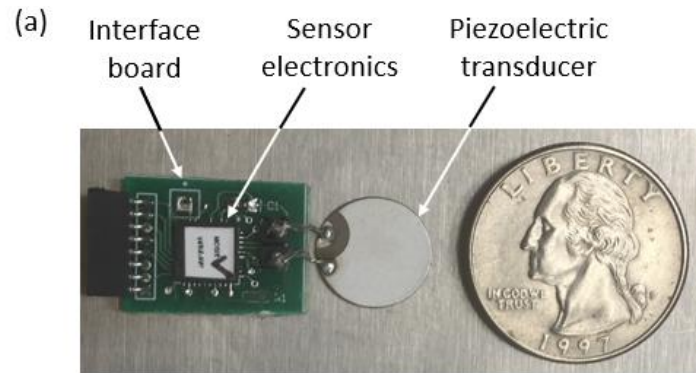
Self-powered, Wireless Sensors



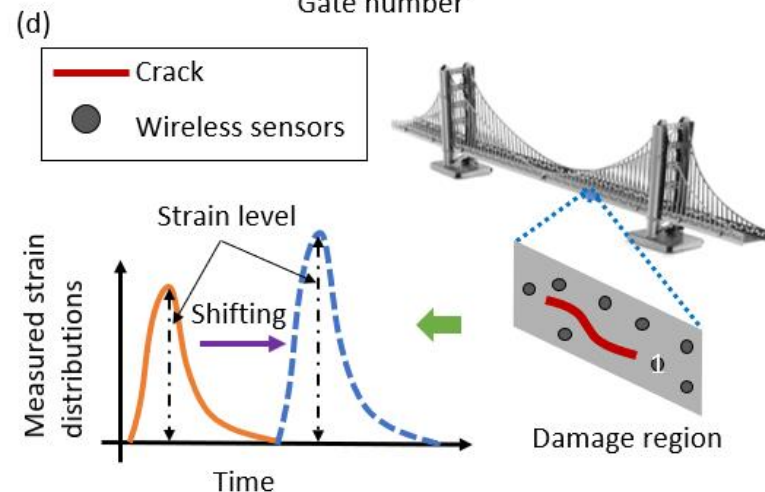
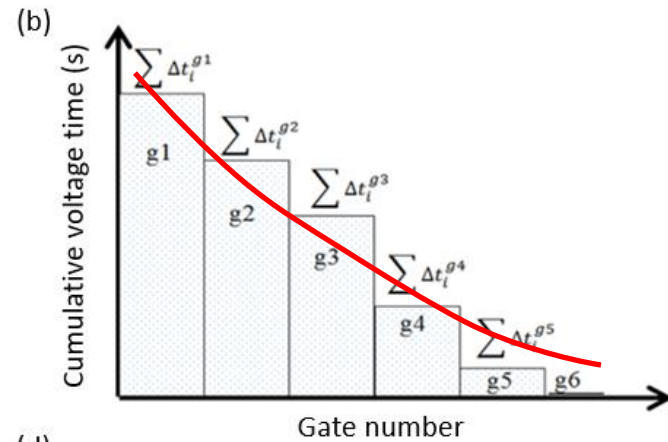
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors

Initial Detection



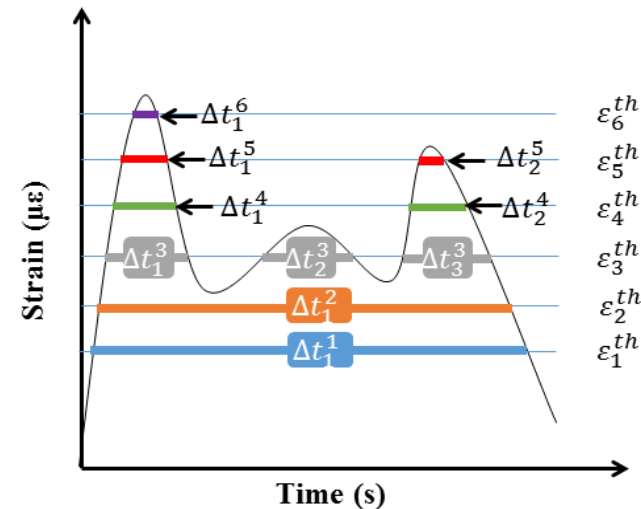
CDF Fitting



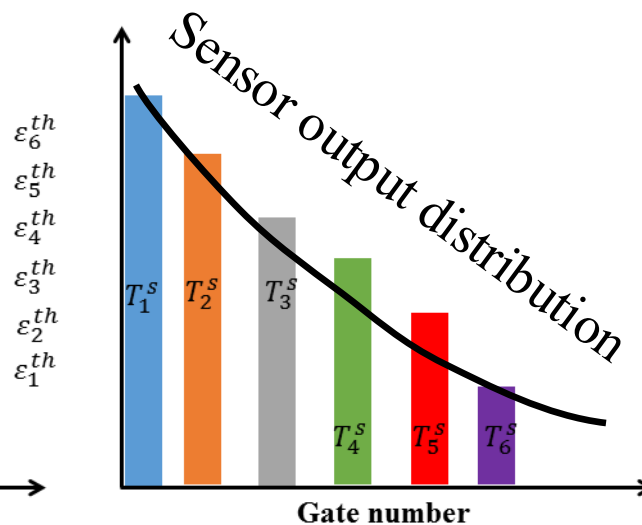
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors

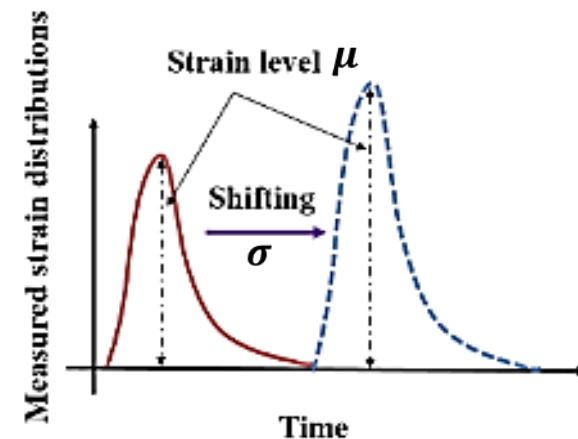
Initial Detection Data



CDF Fitting



Transformed PDF



Level crossing
cumulative time

$$CDF(x) = \frac{\alpha}{2} \left[1 - \operatorname{erf} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right) \right]$$

$$PDF(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(g - \mu)^2}{2\sigma^2}}$$

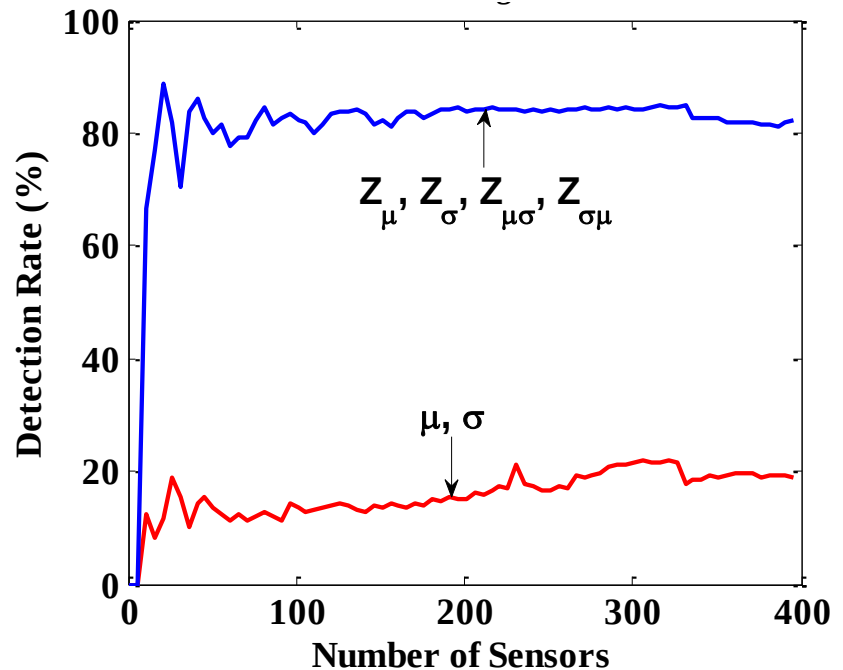
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors

$$\text{Damage} = f(\mu_{ij}, \sigma_{ij}, X_{ij}, Y_{ij})$$

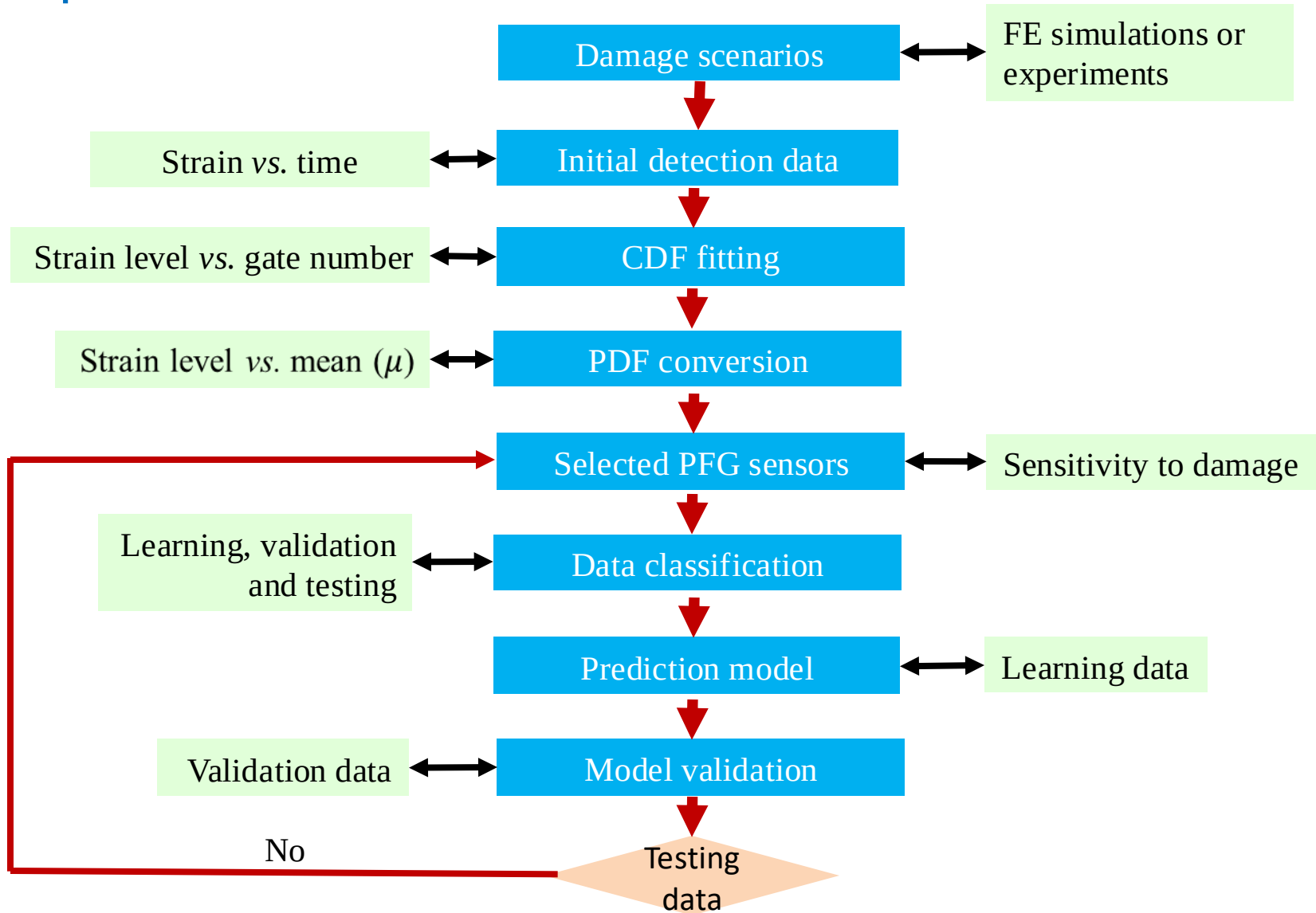
Sensor locations

$$\begin{cases} Z_{\sigma_i}^p = \frac{\sigma_i^p - \sigma_{av}^p}{\sigma_{STD}^p} \\ Z_{\mu_i}^p = \frac{\mu_i^p - \mu_{av}^p}{\mu_{STD}^p} \\ Z_{\sigma\mu_i}^p = \frac{\sigma_i^p - \mu_{av}^p}{\sigma_{av}^p} \\ Z_{\mu\sigma_i}^p = \frac{\mu_i^p - \mu_{av}^p}{\sigma_{av}^p} \end{cases}$$



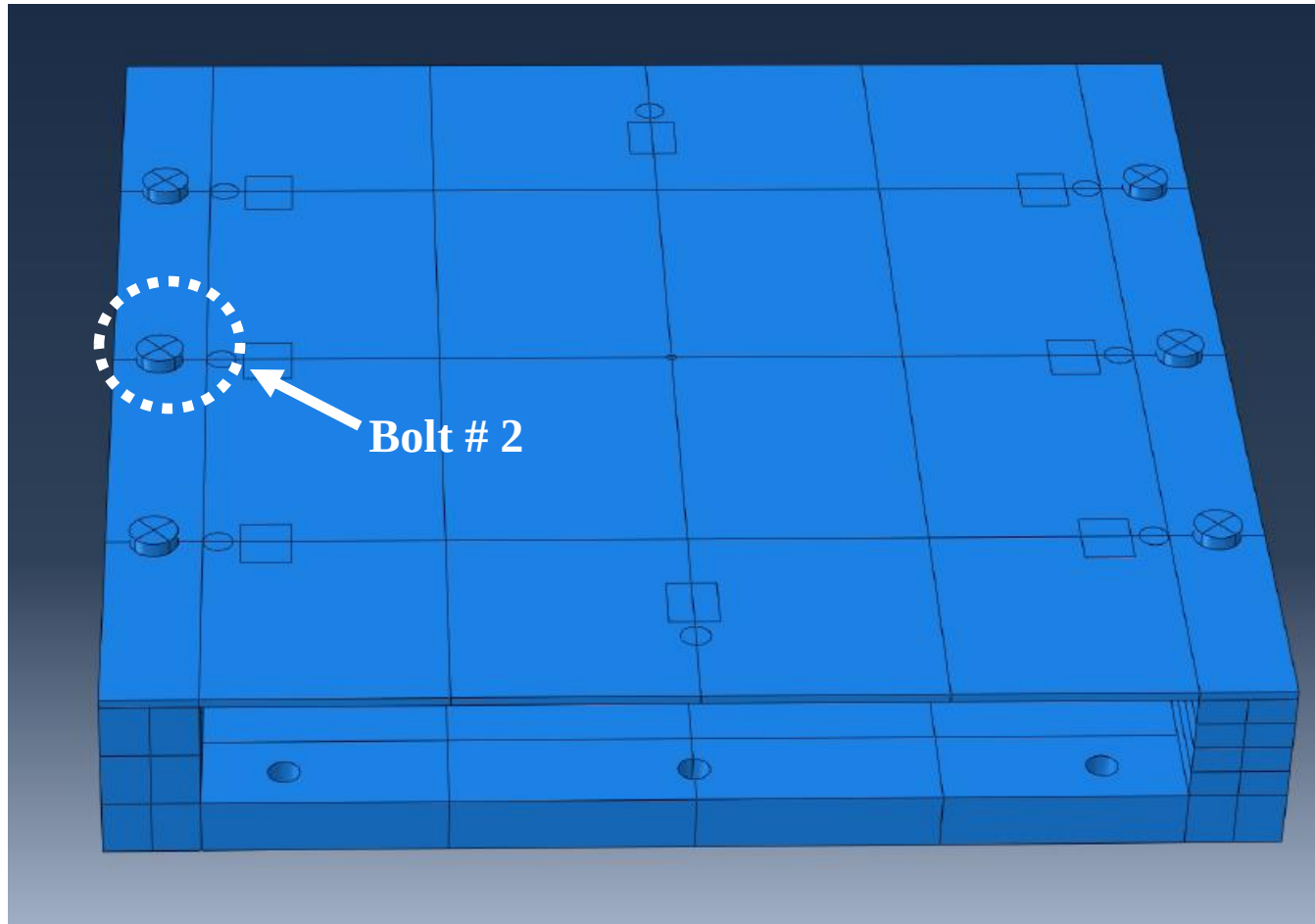
Structural Health Monitoring (SHM)

Self-powered, Wireless Sensors



Structural Health Monitoring (SHM)

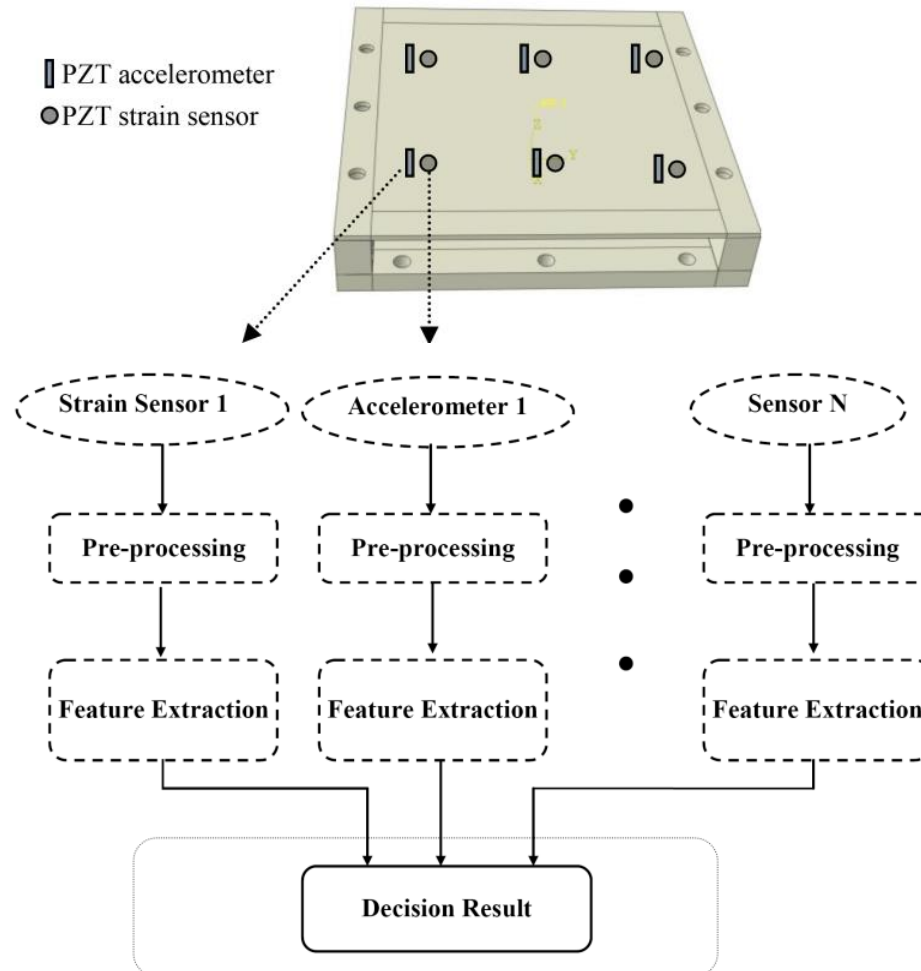
Bolted Steel Plates



Structural Health Monitoring (SHM)

Bolted Steel Plates

Hybrid sensor networks

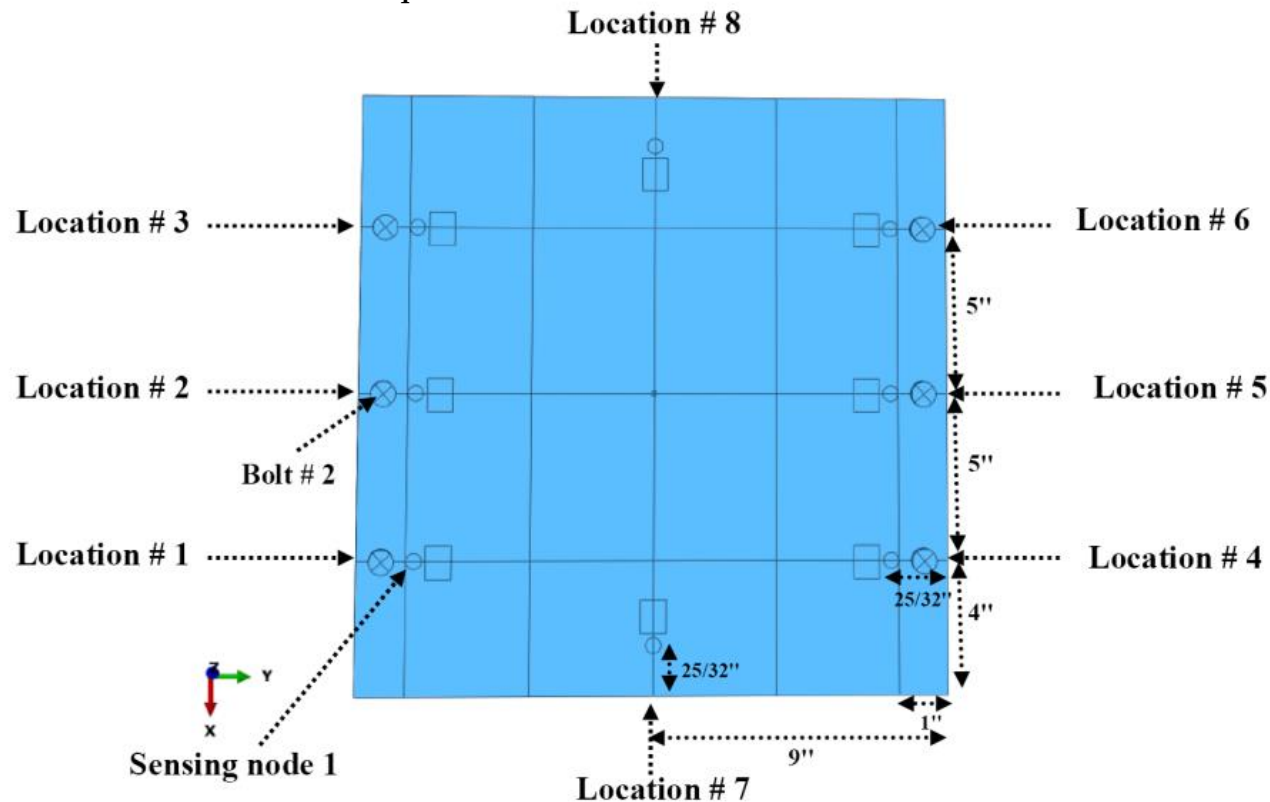


Structural Health Monitoring (SHM)

Bolted Steel Plates

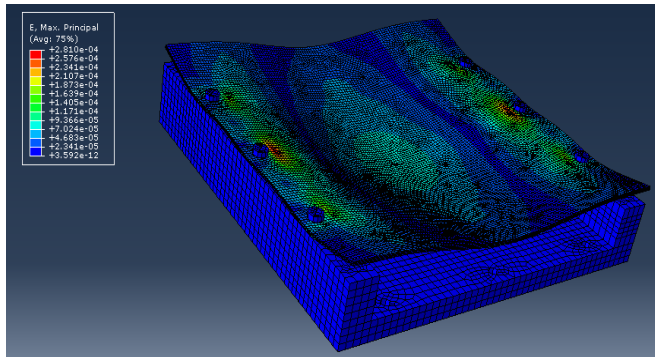
Damage stages

- Undamaged: 100% of nominal torque
- 50% Loosened: 50% of nominal torque
- 90% Loosened: 10% of nominal torque
- Missing Bolt: 0% of nominal torque

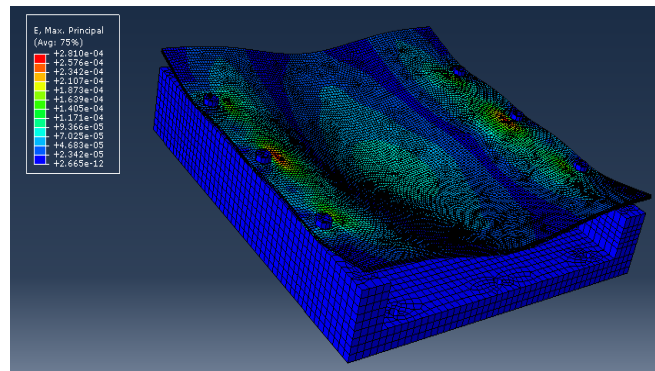
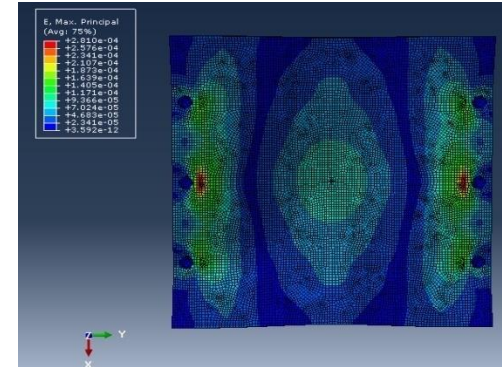


Structural Health Monitoring (SHM)

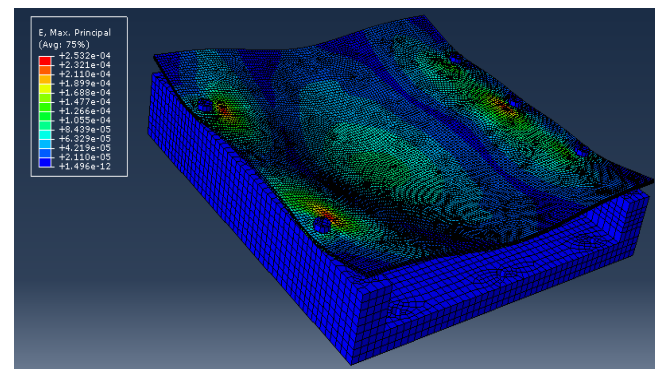
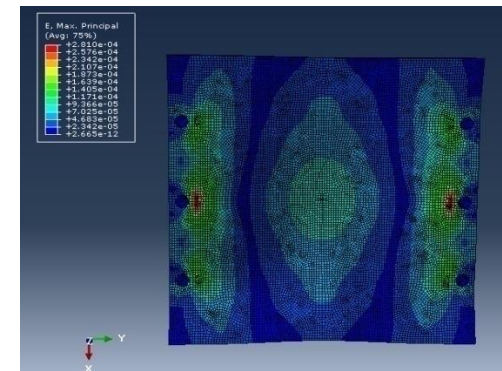
Bolted Steel Plates



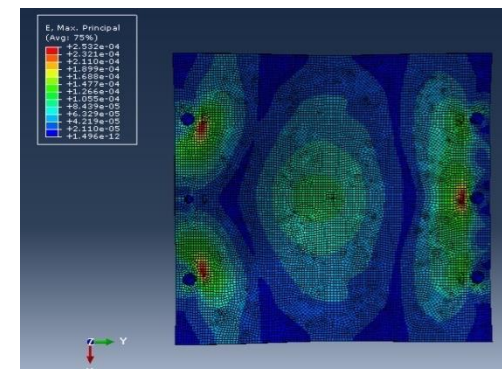
Undamaged



50% Loosened

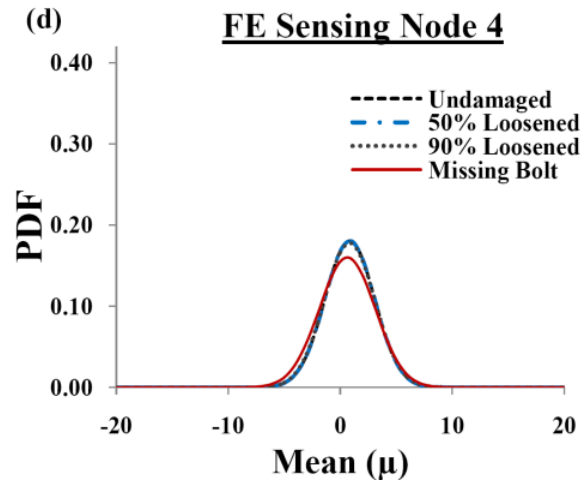
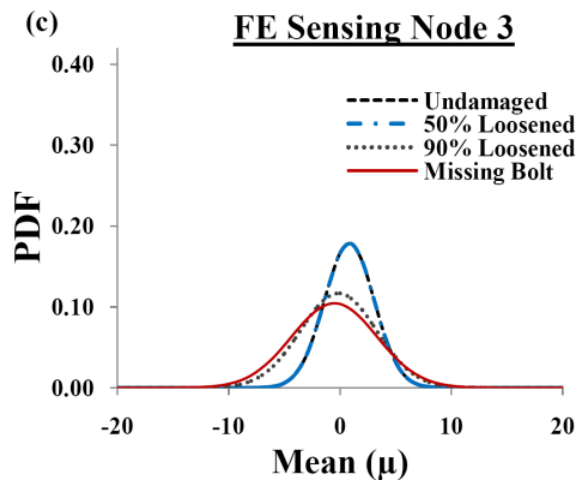
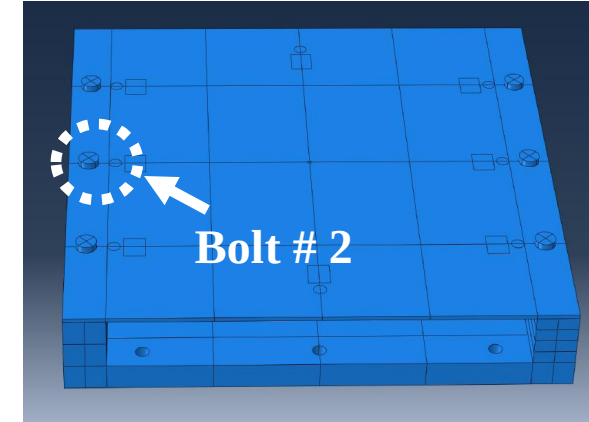
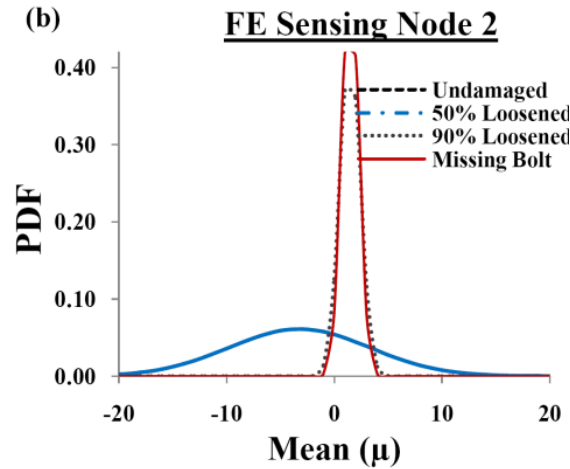
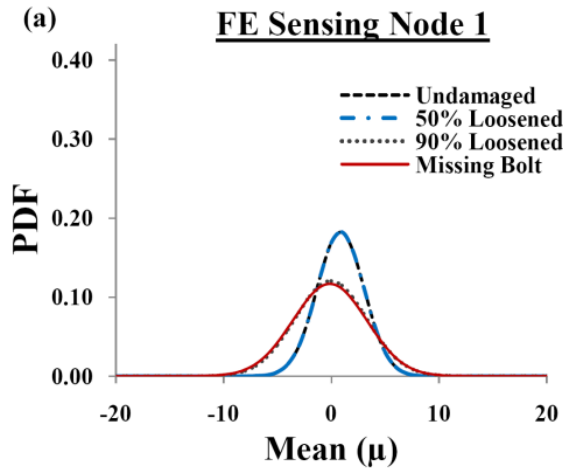


Missing Bolt



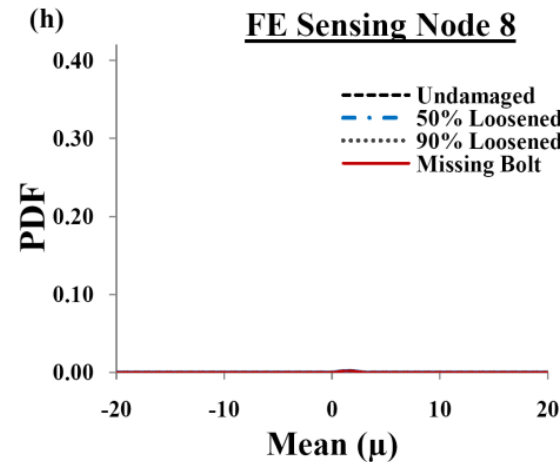
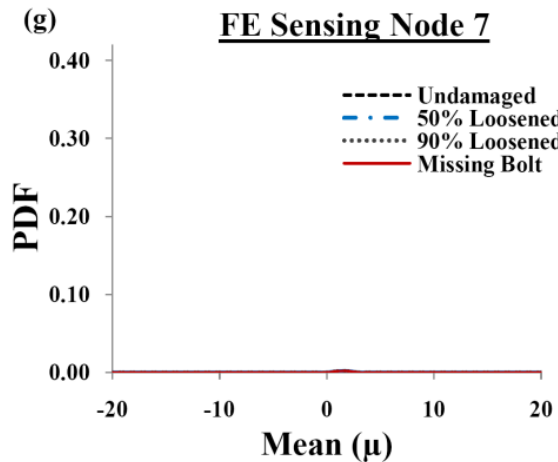
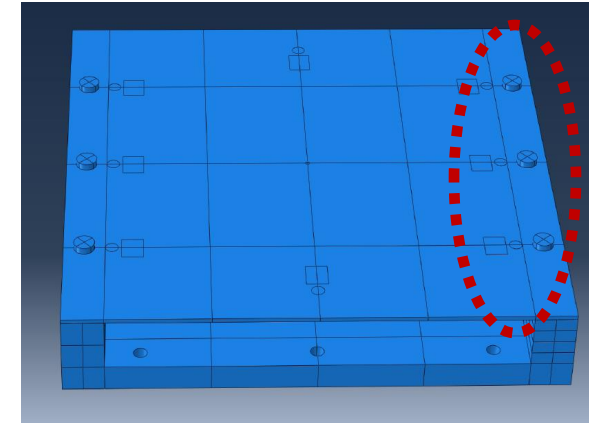
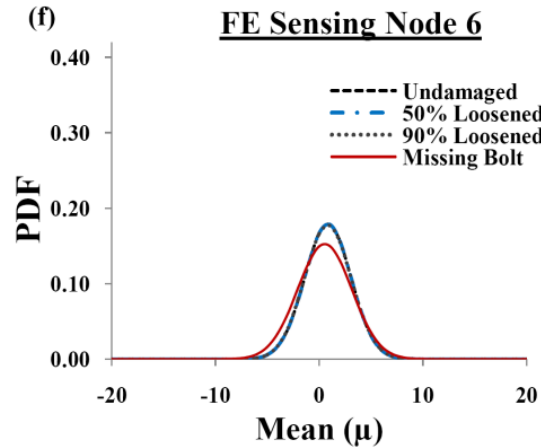
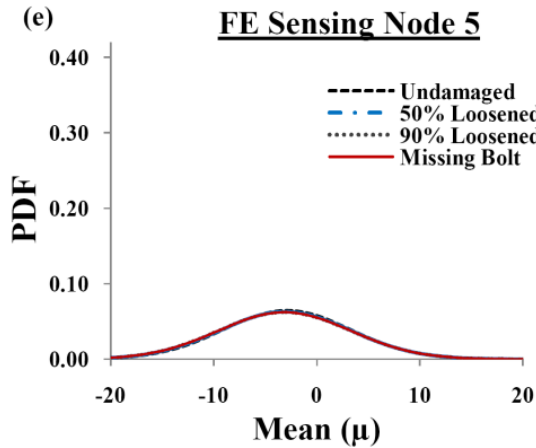
Structural Health Monitoring (SHM)

Bolted Steel Plates



Structural Health Monitoring (SHM)

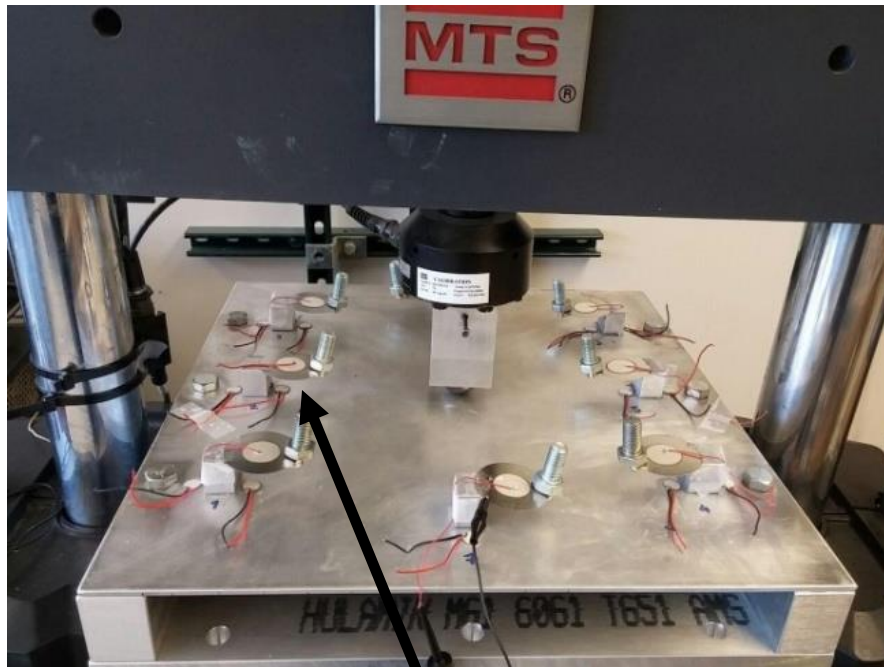
Bolted Steel Plates



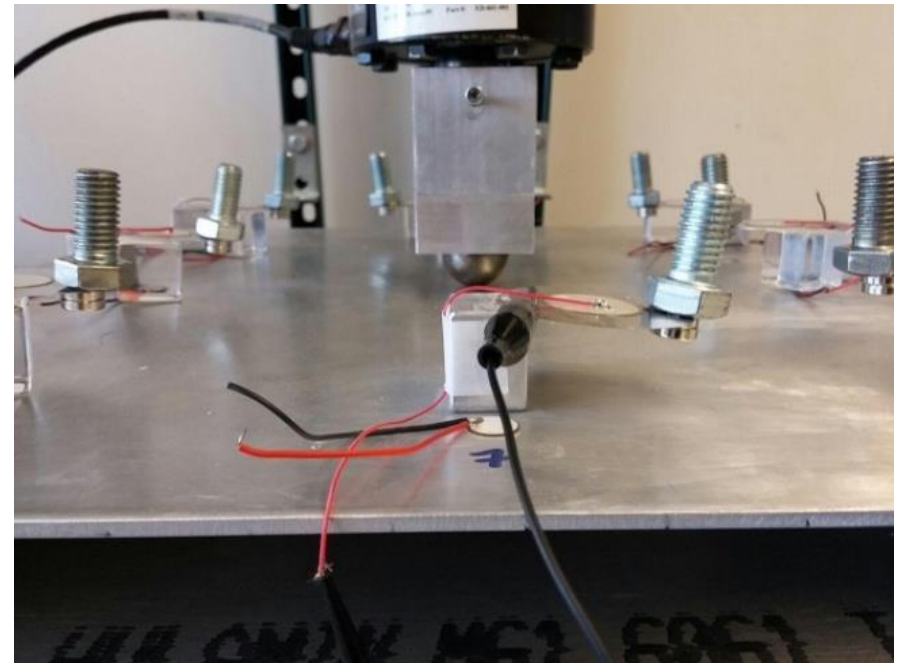
Structural Health Monitoring (SHM)

Bolted Steel Plates

Experimental validation



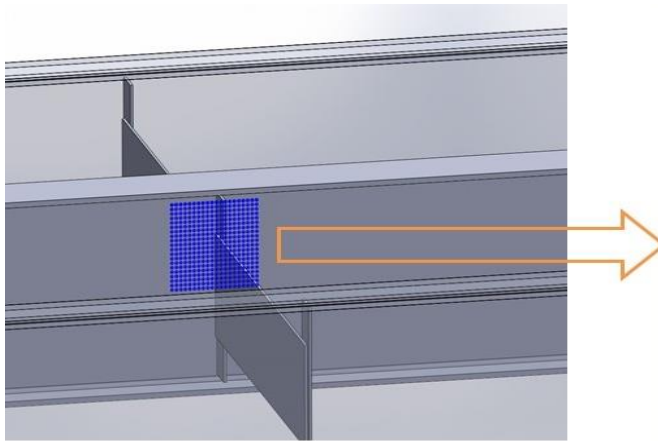
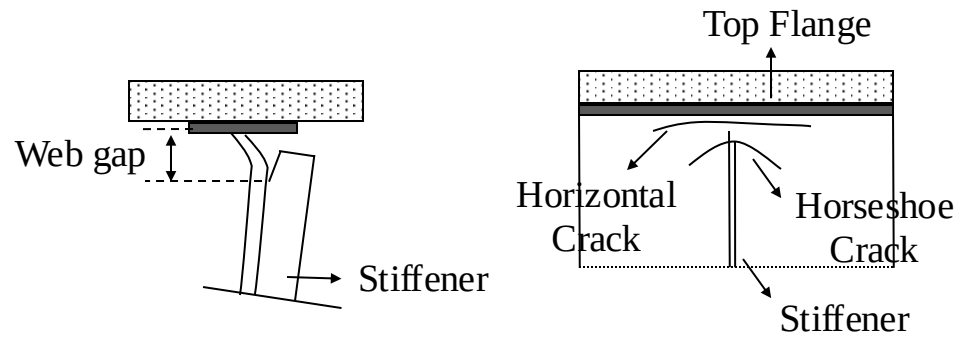
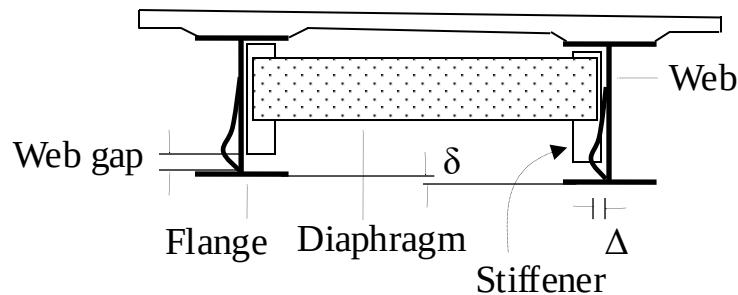
Bolt # 2



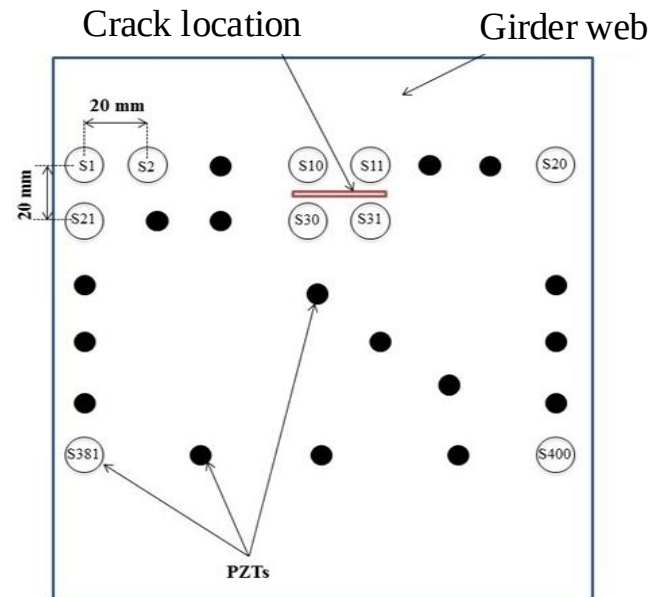
Structural Health Monitoring (SHM)

Steel Concrete Girder

Distortion-induced fatigue cracking



Crack location

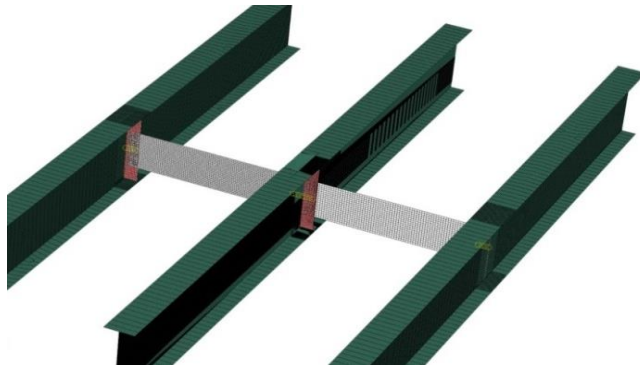


Sensor location

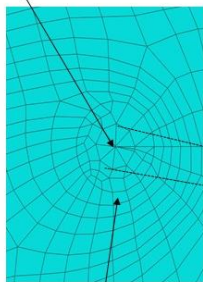
Structural Health Monitoring (SHM)

Steel Concrete Girder

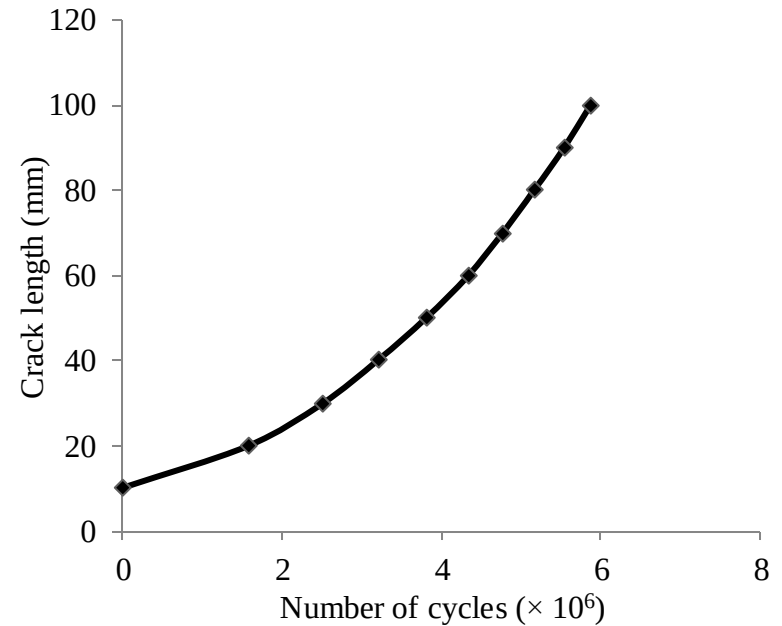
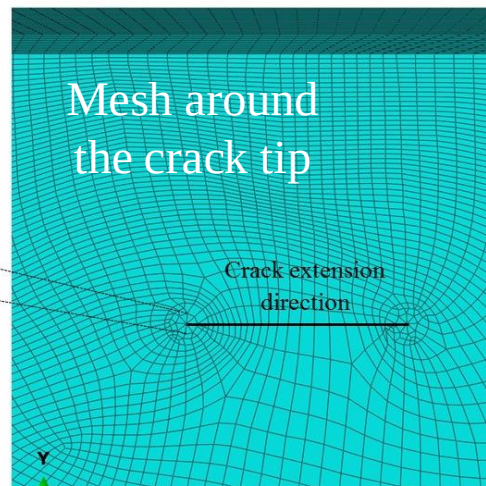
FE Simulations



Crack tip mesh:
Triangular elements
(S3)



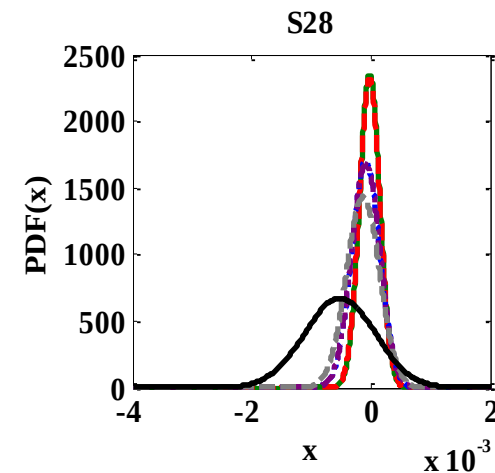
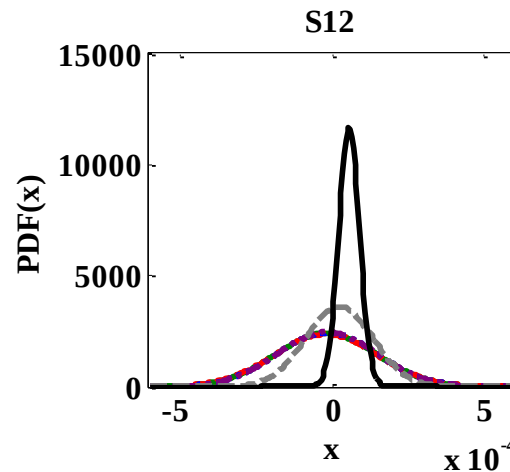
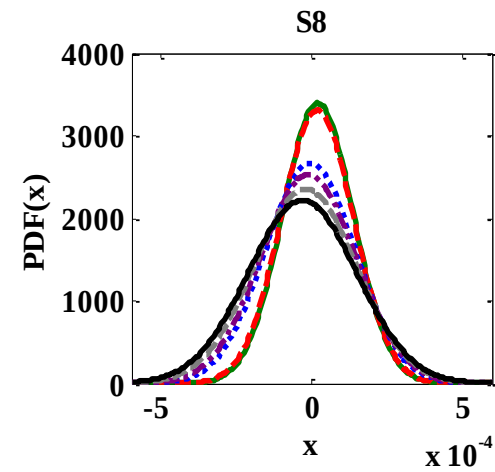
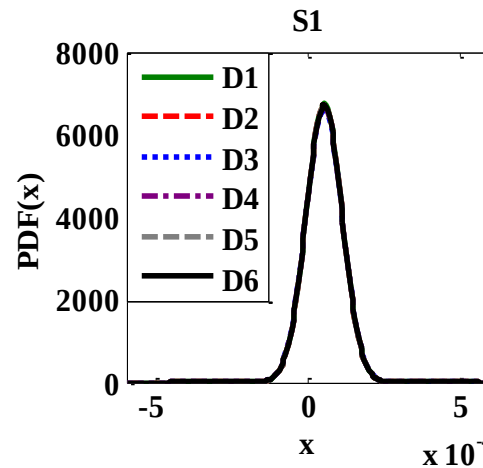
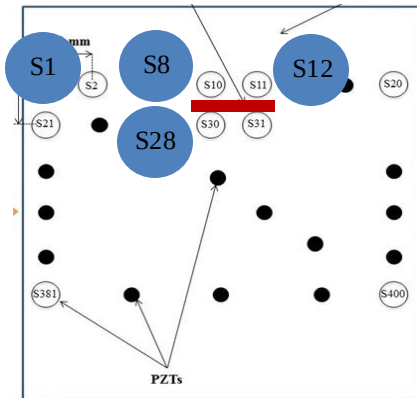
Quadrilateral
elements (S4R)



Structural Health Monitoring (SHM)

Steel Concrete Girder

Initial Detection Data to PDF Function



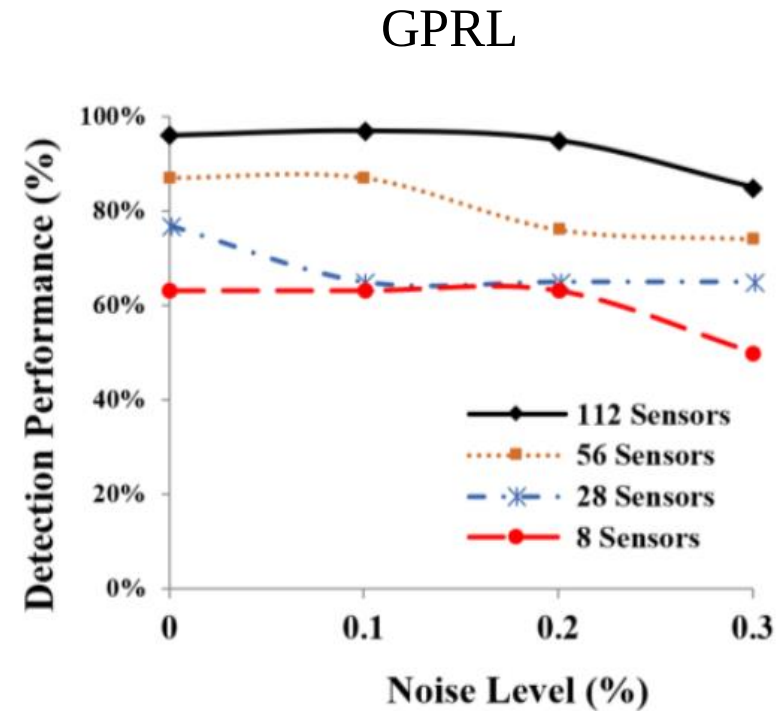
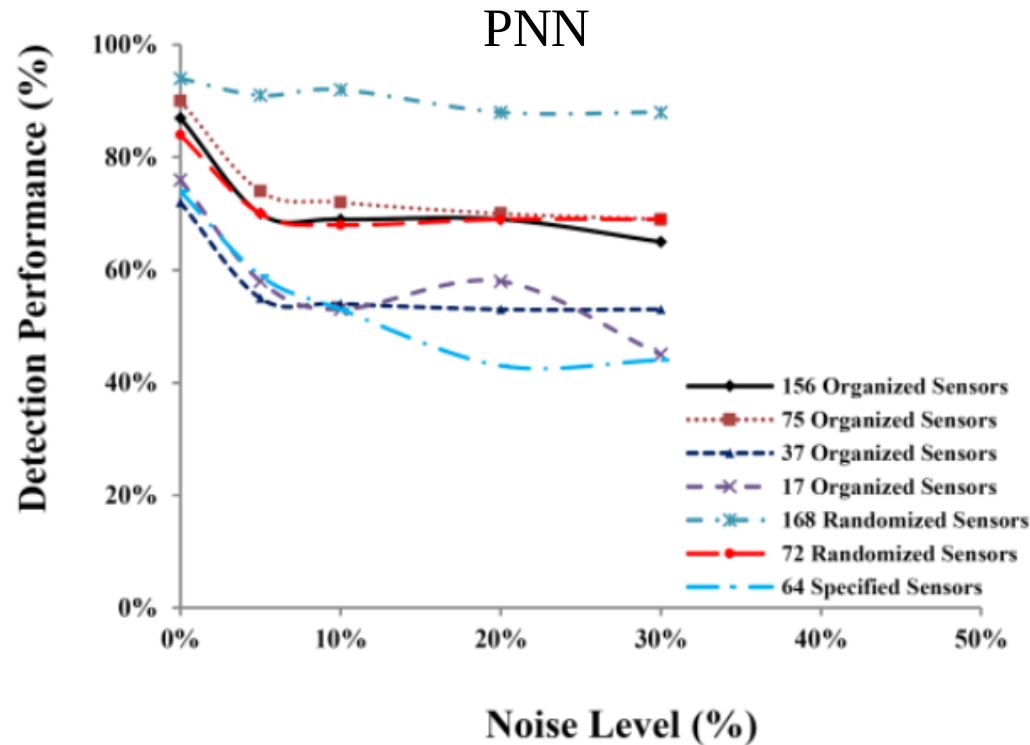
Steel Concrete Girder



Structural Health Monitoring (SHM)

Steel Concrete Girder

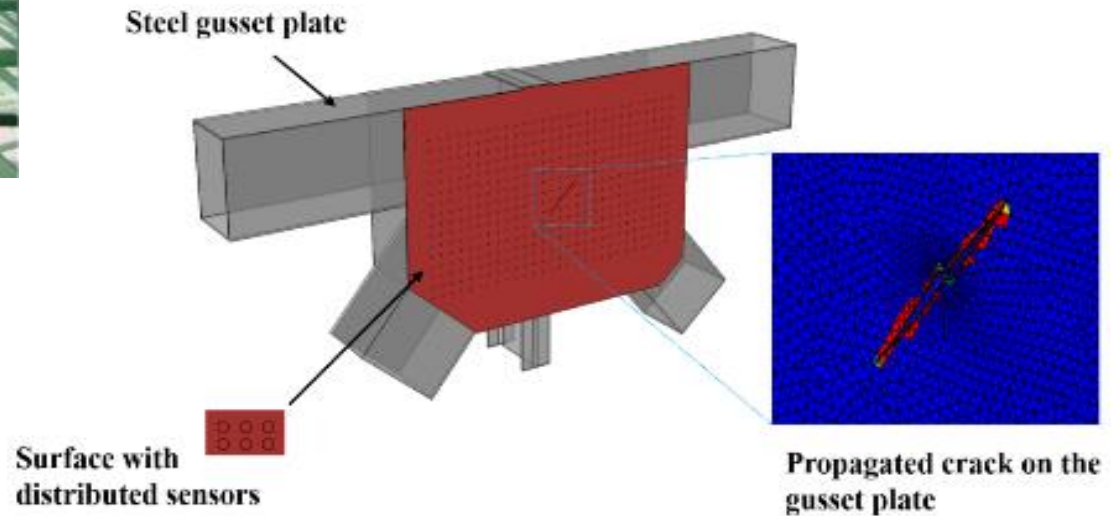
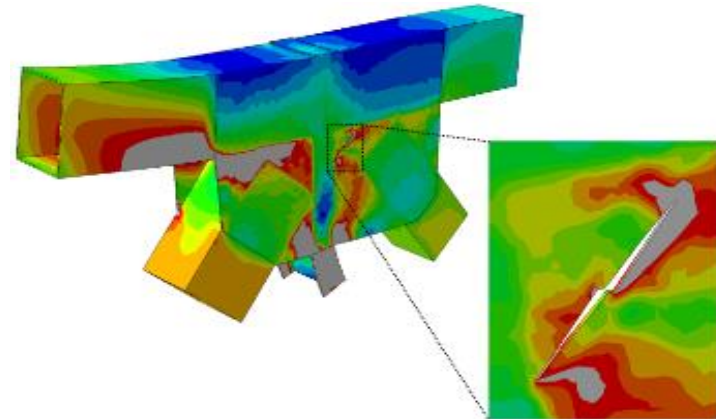
PNN vs. GPRL



Structural Health Monitoring (SHM)

Gusset Plates

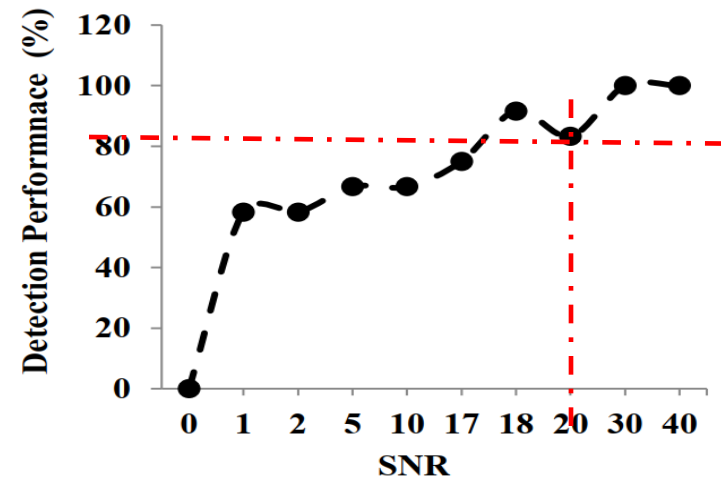
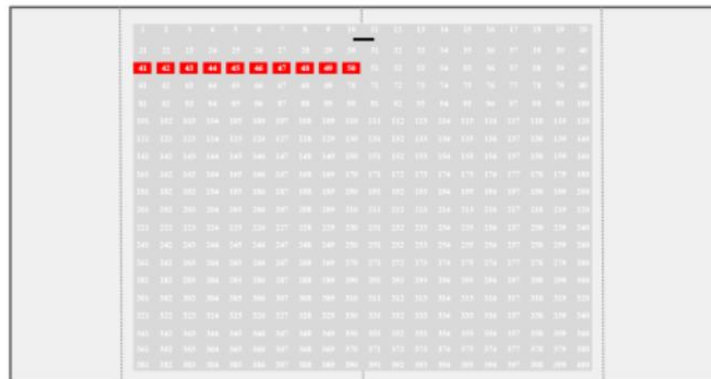
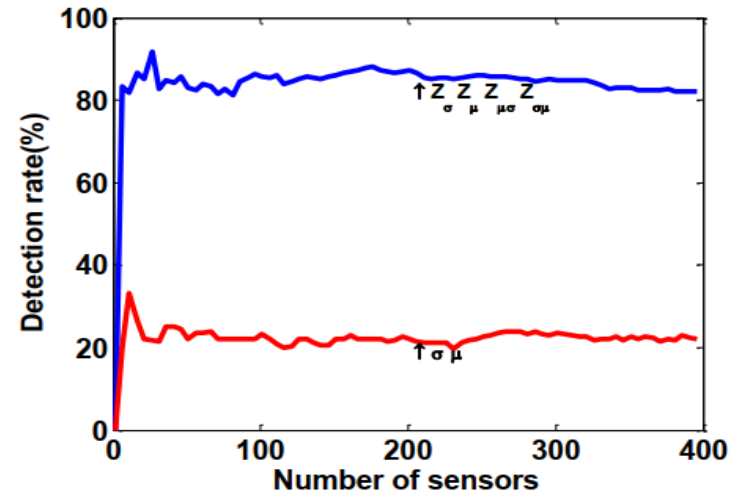
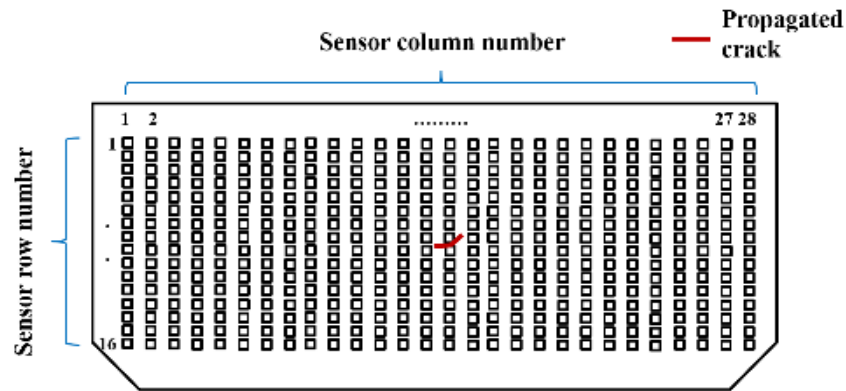
FE Simulations



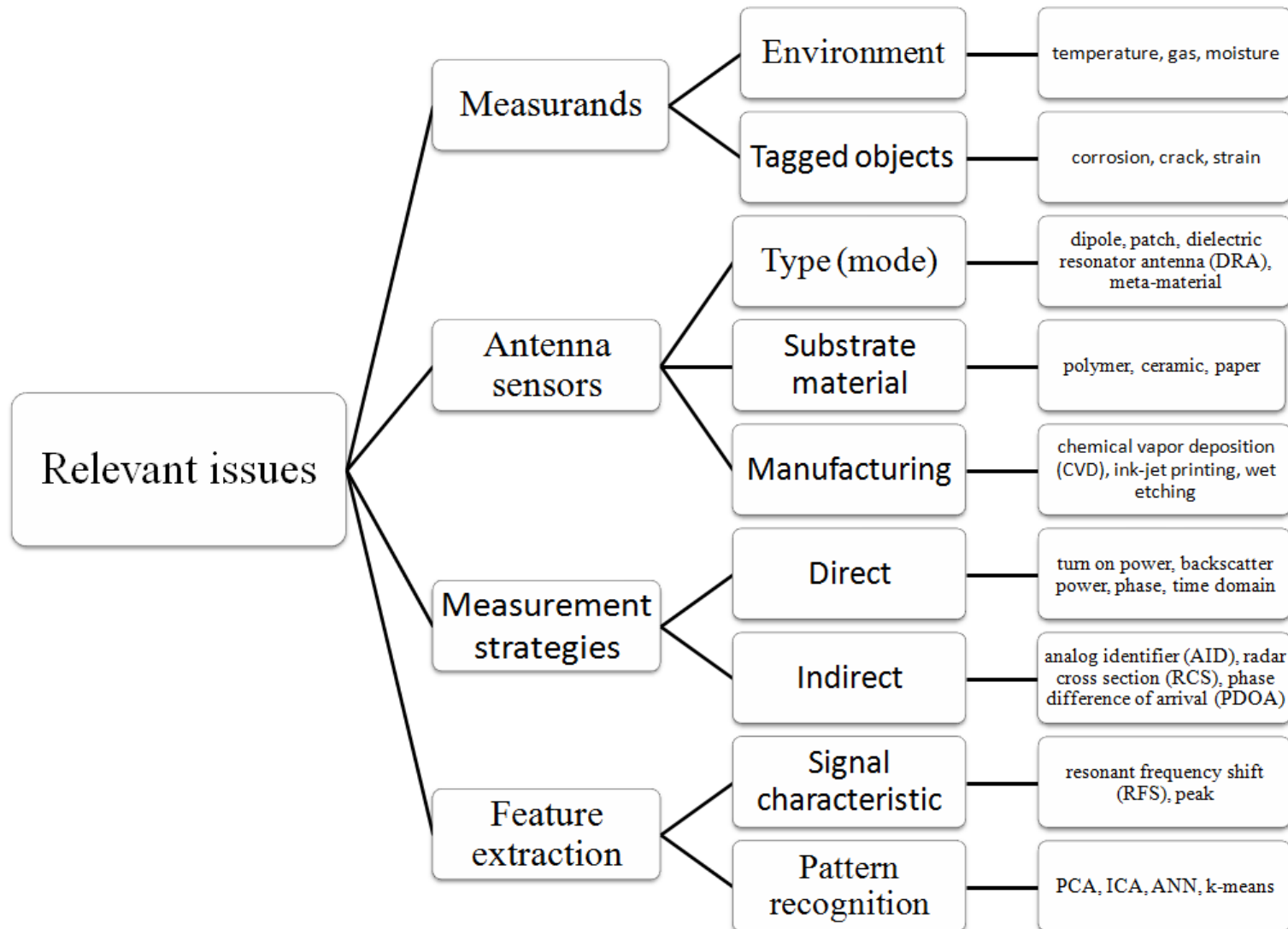
Structural Health Monitoring (SHM)

Gusset Plates

Sensor Selection and Detection Performance

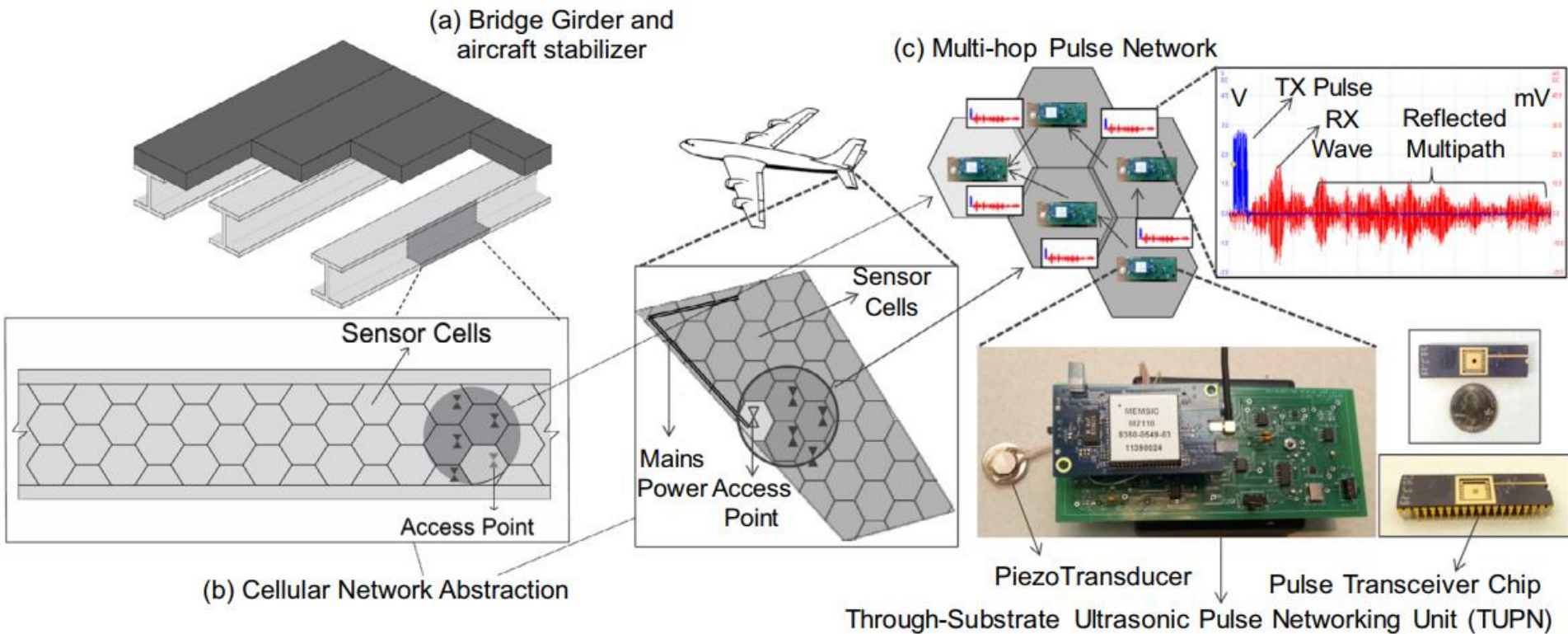


AI-Based Smart SHM



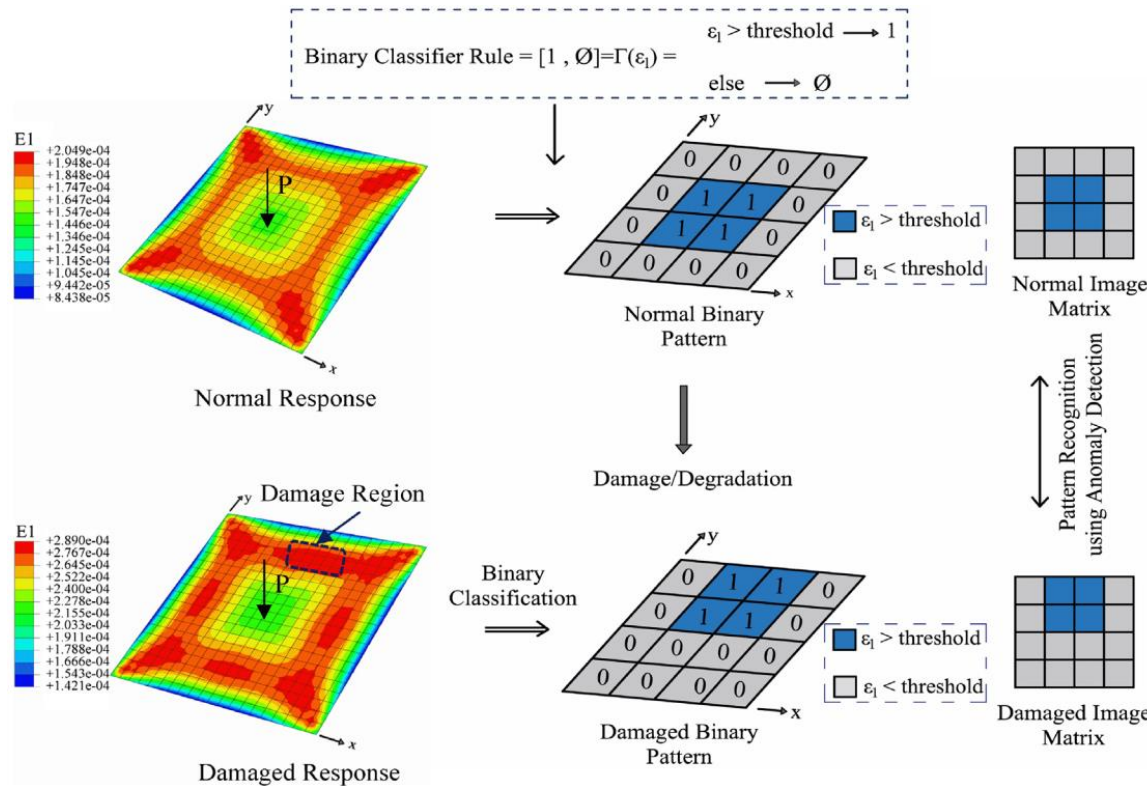
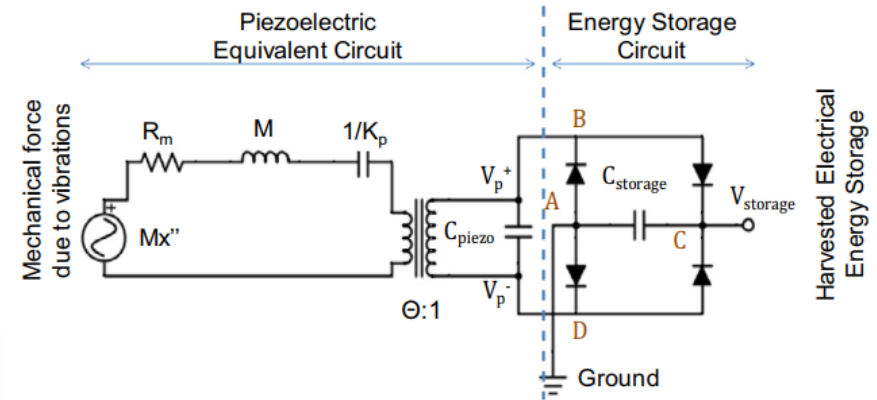
AI-Based Smart SHM

Piezo Sensing and PR



AI-Based Smart SHM

Piezo Sensing and PR



AI-Based Smart SHM

Piezo Sensing and PR

Finite Element Modeling of a Simply Supported Plate for Different Damage Scenarios

Determination the Accumulative Acceleration Response at Sensor Nodes

Introduce an Image-Based Schema

Pattern Recognition (PR) based on Image Data Analysis Techniques
(Proposed PR method using Anomaly Detection)

Identification of Normal and Abnormal Patterns



Determination of the Reliability Factor for Abnormal Patterns

Identified Abnormal Patterns with
High Reliability Factor

Identified Abnormal Patterns with
Low Reliability Factor

Patterns Representing Actual Damage

Patterns Representing Time Delay and Noise

Determination of the Focal Density of Binary Data

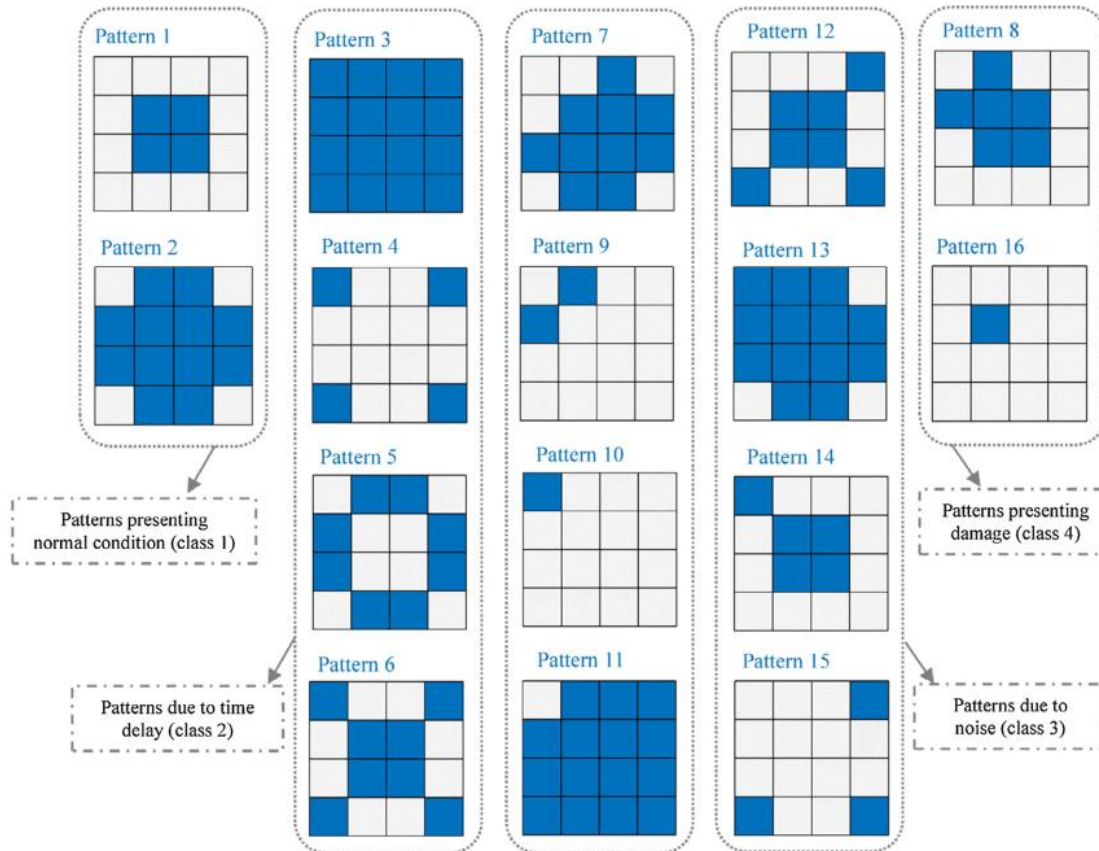
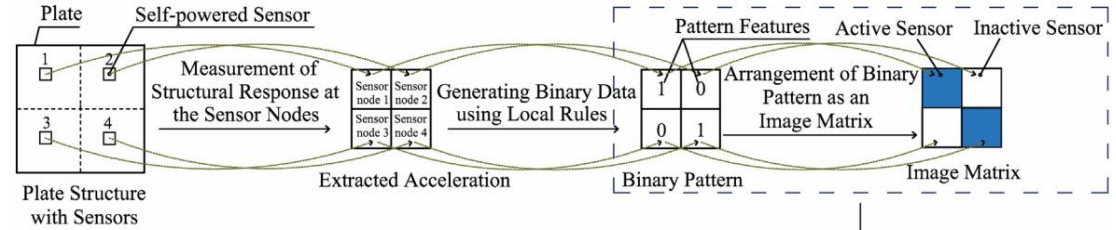
Detection of the Location of Damage

K-Nearest Neighbor Method

Performance Evaluation of localized Damage Detection Strategy

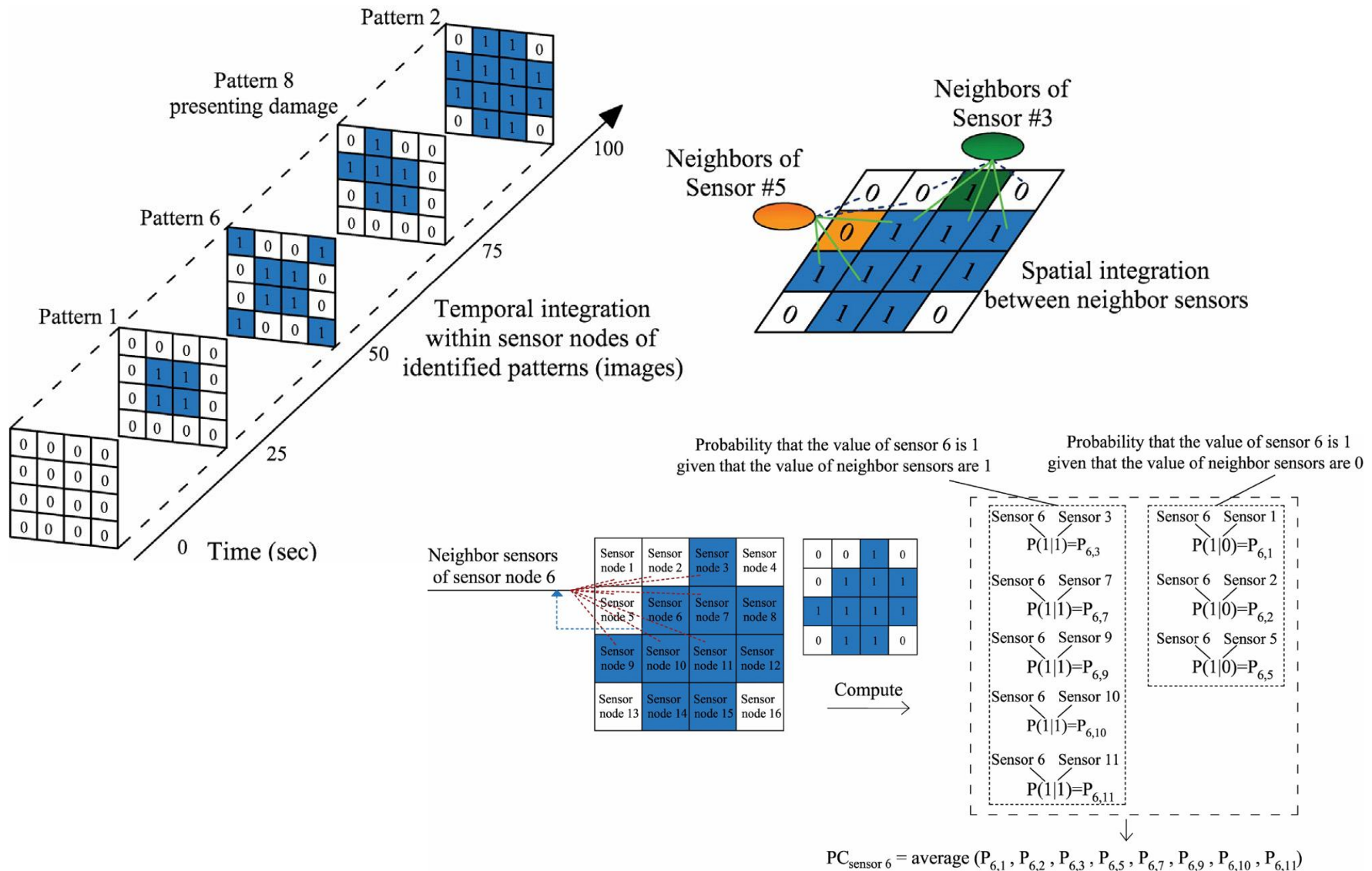
AI-Based Smart SHM

Piezo Sensing and PR



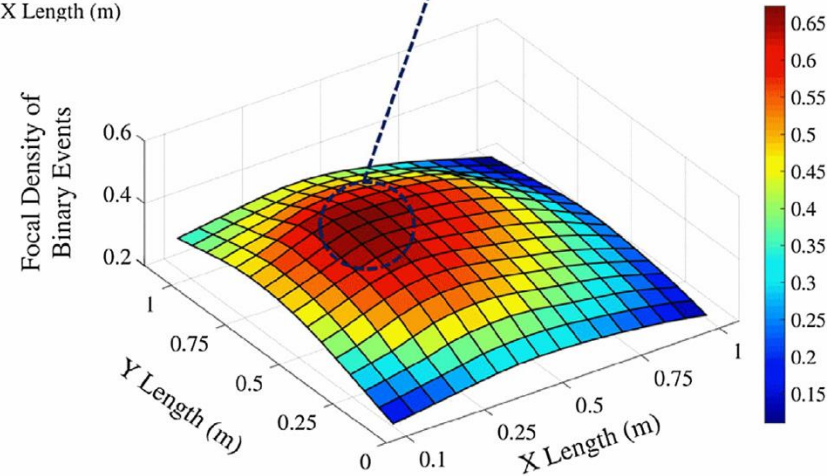
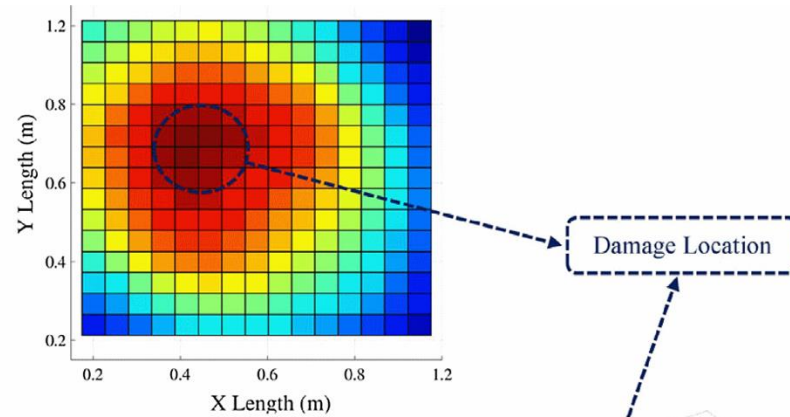
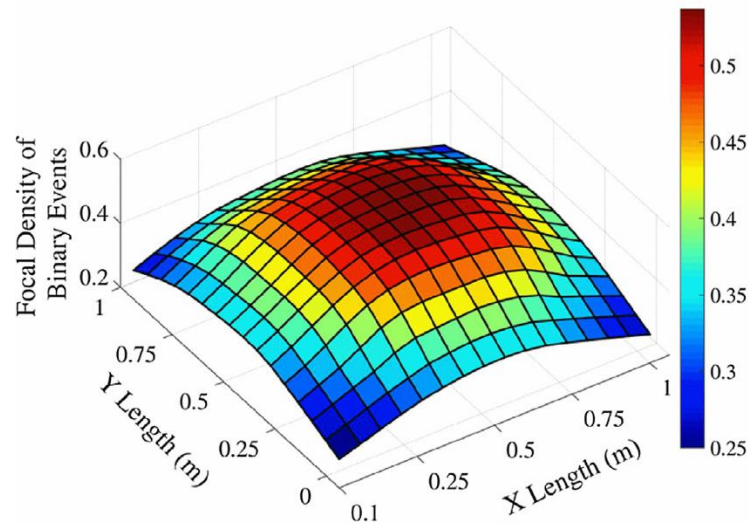
AI-Based Smart SHM

Piezo Sensing and PR



AI-Based Smart SHM

Piezo Sensing and PR



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浙江大学海洋学院
Ocean college, Zhejiang University

